Even such systems may fail and dual equipment is therefore sometimes installed in parallel cells. With the process running on the spare equipment, the failing equipment must be repaired. This requires efficient decontamination both on inside and outside of the equipment, provisions for which must be incorporated in the original design. This design requires dividing the plant in a large number of shielded cells so that the repair workers are protected from the radiation of the functioning plant.

Remote maintenance is more expensive, but may be safer for the personnel and desirable from the standpoint of continuity of operations because equipment replacements can be carried out quickly and interruption of operation is relatively brief. Modern robot technology simplifies such remote maintenance.

21.7. REPROCESSING OF THORIUM FUELS

The thorium-containing fuels of present interest are only those of the kind used in HTGR and HWR; in the future fuel from MSR-like[4] (or other) transmutation devices may become important. In the HTGR fuel elements the fertile ThO_2 and fissile $^{235}UO_2$ (or $^{233}UO_2$) particles are coated differently and embedded in a graphite matrix.

In case of the HTGR, the spent graphite fuel elements are mechanically crushed and burned to eliminate the graphite matrix and pyrolytic carbon coating from the fuel particles. Leaching permits separation of the fissile and the fertile particles because the fissile particles have a silicon carbide coating which remains intact during burning and leaching, while the all-pyrolytic carbon coatings on the fertile particles are burned away, allowing the oxide ash to be dissolved by a leach solution, consisting of HNO_3 and F^-. The solution is clarified and adjusted to proper acidity for solvent extraction (paragraph (i) below). The undissolved residue resulting from clarification is dried and classified for further treatment (paragraph (ii) below). The burner off-gas streams are passed through several stages of filtration, scrubbing, and chemical reaction to remove the entrained and volatile fission products, as well as ^{14}C containing CO_2, prior to atmospheric discharge.

In case of oxide fuel, it has to be cut and dissolved in a way similar to that described for uranium fuels.

(i) The *acid Thorex solvent extraction process* is used to purify and to separate the ^{233}U and the thorium. Three solvent extraction cycles are used. In the first, the uranium and thorium are coextracted by 30% tributyl phosphate (TBP) from $\sim$5 M HNO_3 and then stripped into an aqueous phase. In the second cycle, the uranium and thorium are separated by controlling the extraction conditions using $\leq$1 M HNO_3. The uranium is extracted and processed by an additional solvent extraction cycle for final purification, while the thorium remains in the aqueous raffinate stream. Following

[4] MSR = Molten Salt Reactor.

concentration and assay, the uranium is ready for fuel refabrication. The partially decontaminated thorium is concentrated and stored.

(ii) The separated silicon-carbide coated fissile particles from the head end process for HTGR fuel are mechanically crushed to expose the fuel and are burned to remove carbon and oxidize the fuel material: the ash is leached to separate the fuel and fission products from the coating waste. The ^{235}U is separated from the fission products by solvent extraction using a PUREX flowsheet. An organic solvent containing only 3–5% TBP is used in order to avoid criticality problems. A reductant such as Fe(II) sulfamate is added to the feed in order to force the small amount of plutonium present into the aqueous raffinate stream. The waste streams are treated as in the PUREX process.

The process is plagued by both chemical and nuclear difficulties. The decay chain $^{233}Th \rightarrow {}^{233}U$ forms 27 d half-life ^{233}Pa. For a complete decay of all ^{233}Pa to ^{233}U, the spent fuel elements must be cooled for about a year. A still considerable amount of longlived ^{231}Pa ($t_{½}$ 32 760 y) is present in the spent fuel (about 1/2000 of the amount of ^{233}U); protactinium complicates the reprocessing chemistry and constitutes an important long-lived waste hazard.

The ^{233}U isolated contains some ^{232}U formed through reactions indicated in Figure 20.3. Since the half-life of ^{232}U is rather short and its decay products even shorter lived, a considerable γ-activity will grow in with the ^{233}U stock, complicating its handling. Since the first decay product is ^{228}Th, some ^{228}Th forms in the fuel elements, making the ^{232}Th contaminated by this isotope. Thus neither the ^{233}U nor the ^{232}Th produced can be free from γ-activity.

21.8. WASTES STREAMS FROM REPROCESSING

Fuel reprocessing generates a large variety of wastes which can be classified in categories according to activity (low, medium and high), physical state (gas, liquid or solid package), or decay characteristics (shortlived, longlived), each treated separately. The amounts of different categories are given in Table 21.8; these wastes are treated internally at the plant and not released to the environment. Because reprocessing plants vary considerably the amounts of wastes produced also differs, especially for the liquid medium and low level categories. In this section we briefly discuss the various waste streams appearing at the plant and the treatment methods.

21.8.1. Gaseous wastes

Gaseous wastes originate mainly from the chopping and dissolution operations. In current practice the volatile radionuclides are discharged to the stack after scrubbing with sodium hydroxide and filtration through a special zeolite or charcoal filter. The hydroxide scrubbing removes the acidic nitrous oxides which pass through the